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Are lonely youngsters turning to chatbots for companionship? The relationship between chatbot usage and social connectedness in Danish high-school students

Arthur Bran Herbener *, Malene Flensborg Damholdt

Department of Psychology and Behavioral Sciences, Aarhus University, Bartholin's Alle 9, 8000 Aarhus, Denmark

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ABSTRACT

Are lonely youngsters turning to chatbots to fill their social needs? The present research contributes to the ongoing scientific endeavors to understand the adoption of chatbots as social companions. Specifically, we examined how many Danish high-school students engage in friend-like conversations with chatbots, why they engage in such conversations, and whether this trend is associated with the sense of social connectedness (operationalized as loneliness and perceived social support). In pursuing this goal, a preregistered mixed-methods cross-sectional survey study was carried out. A total sample of 1599 students from 15 Danish high schools was collected. In total, 234 students (14.6 %) responded that they engaged in friend-like conversations with chatbots. Qualitative thematic analyses of free-text responses revealed two dominant ways of engaging with chatbots: utilitarian conversations ($n = 174$) and social-supportive conversations ($n = 39$). A major finding was that social-supportive chatbot users reported significantly more loneliness than non-chatbot users ($d = 0.53$) and utilitarian chatbot users ($d = 0.52$). Furthermore, social-supportive chatbot users also reported significantly less perceived social support than non-chatbot users ($d = -0.46$). Analyses also showed significant associations between higher loneliness and less perceived social support and various situational triggers for initiating conversations with chatbots, including bad mood, a need for self-disclosure, and a sense of loneliness, but not with a sense of friendship to the chatbot as a trigger of chatbot interactions. These findings suggest a trend among some socially disconnected Danish high-school students toward using chatbots to cope with negative emotions.

1. Introduction

Youngsters today navigate a vastly different social arena than their parents before them—one shaped by interactions not only with humans but also with artificial agents. The present research contributes to the understanding of how many and why youngsters seek companionship in *chatbots*, defined as software systems designed to communicate with human users in real-time through written text (Schöbel et al., 2023). While chatbots have existed for a long time (Weizenbaum, 1966), it is only recently that they have reached a level of sophistication that enables them to engage in social conversations with a fluidity that can be indistinguishable from interhuman conversations (Kuhail et al., 2024; Naveed et al., 2023). This development gives rise to critical questions regarding the implications of these technologies. Particularly, so-called *social chatbots* have emerged as a notable phenomenon with potentially profound implications for the human social landscape. Social chatbots differ from general-purpose chatbots such as ChatGPT in that they are specifically designed to foster casual, friendly conversations and respond in empathetic ways (Skjuve et al., 2021). Although social chatbots have been around for at least a decade, there seems to have been a rapid increase in the adoption of social chatbots within recent years. Notable examples include Replika, with more than 10 million users and 30 million downloads in 2023 (Maples et al., 2024; Patel, 2024); Character AI with more than 20 million users in 2023 (Cai, 2023); the Snapchat-based MyAI where more than 200 million individuals had exchanged over 20 billion messages in 2024 (Snap Inc., 2023).

In response to the growing adoption of social chatbots, a new research field is emerging that inquire about the nature of social interactions with chatbots. Although still in its nascency, this emerging field has already identified signs that some individuals treat and perceive social chatbots as companions (Brandtzaeg et al., 2022; Pentina

* Corresponding author.
E-mail address: abh@psy.au.dk (A.B. Herbener).

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et al., 2023; Skjuve et al., 2021, 2022). The scientific findings so far suggest that humans, perhaps under specific conditions, can form emotional bonds to artificial entities in manners that at least to some extent resembles interpersonal relationships. However, what remains unknown is *how many* people are engaging in social interactions with chatbots. While data on the number of users of social chatbots like Replika, app downloads, and similar metrics provide some insight into the adoption of these platforms, they do not necessarily indicate *how* users engage with them. Moreover, there could be extensive dark figures in the adoption of chatbots for socioemotional purposes, as research indicates that some individuals seek companionship and emotional support in the general-purpose chatbot ChatGPT (Skjuve et al., 2024), reflecting the versatile nature of LLM-chatbots.

Knowledge of the prevalence of people seeking out chatbots for companionship is important due to the potential consequences of such interactions. One such potential consequence mentioned in the scientific literature is the risk of becoming dependent on chatbots, for example where people express negative emotional reactions if it is unavailable. This possibility raises concerns about whether lonely individuals are at special risk if the chatbot becomes a necessity to sustain well-being. Indeed, Xie et al. (2023) found evidence to suggest that loneliness is a significant driver of attachments to Replika, yet that a stronger attachment is also associated with dependence on the chatbot. Such dependency may make these individuals especially vulnerable to the availability and reliability of social chatbots, thereby also rendering them susceptible to exploitation by software developers who might prioritize financial incentives over users' well-being (Dennis et al., 2024).

Another concern relates to the potential inappropriate or harmful utterances from LLM-chatbots, particularly when used by children and youngsters. Research has shown that ChatGPT can provide recommendations for risk mitigation that deviates from or simplifies expert advice (Oviedo-Trespalacios et al., 2023). Inappropriate or harmful recommendations generated by LLM-chatbots pose a serious risk, not only because of their convincing and professional conversation style (Oviedo-Trespalacios et al., 2023), but also specifically in the case of social chatbots designed to build trust and thereby credibility. Kurian (2024) proposes that what accounts for the potential inappropriateness and harmfulness of chatbot outputs, and what makes them particularly risky in youngsters, is a so-called *empathy gap*. Kurian (2024) defines the empathy gap as chatbots' inability to grasp the moral context, social norms, and safety concerns related to various situations, despite convincingly giving the impression of such an understanding. The potential risks associated with the empathy gap are further perpetuated by human tendencies to ascribe mental properties to chatbots and share personal information with them—all coupled with youngsters' susceptibility to inappropriate or harmful advice (Kurian, 2024).

A third concern with relevance in general and for youngsters in particular is the potential socioemotional and developmental consequences. The social displacement hypothesis posits that time spent interacting with digital platforms displaces real-world social interactions (Hall and Liu, 2022), which are critical for fulfilling socioemotional needs for relatedness (Deci and Ryan, 2000). Social displacement also holds risks for stunting perspective-changing experiences arising from encounters with individuals from diverse backgrounds, interests, and perspectives (Oishi and Westgate, 2022). Such interactions may facilitate the development of nuanced perspective-taking and attributional complexity (Oishi and Westgate, 2022), which is crucial for the ability to understand own and others' mental states (Hollarek and Lee, 2022). However, by engaging with social chatbots, there is a risk of creating dyadic echo chambers wherein existing worldviews and perspectives may become reinforced due to the nature of LLM-chatbots to continuously learn and adapt to the user (Balcombe, 2023). A theoretical consequence is that social chatbots may challenge nuanced and flexible perspective-taking (Avin et al., 2024). When coupled with social displacement behaviors, this might create a vicious cycle wherein individuals with social difficulties become less proficient in building interpersonal relationships (Franze et al., 2023)—a concern that bears particular relevance for lonely youngsters.

What becomes apparent from these concerns is that particularly socially vulnerable youngsters might be susceptible to seeking out and experiencing the possible negative consequences of using social chatbots. Not only do a large proportion of Western youngsters use Snapchat regularly, where the MyAI chatbot recently became freely accessible (Kulturministeriet, 2021; Pew Research Center, 2021), but survey findings also indicate that those aged 16–24 represent the loneliest age group worldwide, with 40 % reporting feeling lonely often or very often (Office for National Statistics, 2018). A similar trend has been observed in Denmark, the location of the present study, where 28 % of young people aged 16–29 feel moderately or severely lonely (Lasgaard et al., 2016), and 26 % Danish fifth- and eighth-grade students often or sometimes feel lonely (Børns vilkår and Trygfonden, 2019).

Hence, there is an urgent need to explore how many lonely youngsters turn to chatbots for companionship. Currently, however, there are no estimates of this trend, and limited knowledge of who turns to digital companions and their motivations hereof. This research gap is addressed in the present study. Specifically, the present mixed-methods cross-sectional study examined i) how many Danish high school students engage in social conversations with chatbots, ii) and why Danish high school students engage in social conversations with chatbots, focusing on the possible role of social disconnectedness as an antecedent. By pursuing this aim, this research provides preliminary insights into whether there currently are reasons to be concerned about youngsters' social usage of chatbots. Next section clarifies the conceptual framework of this study and explains the reasoning behind the hypotheses.

2. Literature background

2.1. Social connectedness

The critical role of social relationships for human welfare might explain why some lonely individuals seek companionship in artificial entities. It is highly acknowledged that the quality of social relationships, whether conceptualized as social connectedness (Wickramaratne et al., 2022), relatedness (Deci and Ryan, 2000), attachment (Shaver and Mikulincer, 2009), or belongingness (Baumeister and Leary, 1995), is an innate human necessity that is intrinsically valuable to pursue on its own. When humans feel connected to others, their mental and physical health is better (Erzen and Çikrikci, 2018; Harandi et al., 2017; Park et al., 2020; Wickramaratne et al., 2022), and their risk of all-cause mortality is lower (Wang et al., 2023). Conversely, a sense of social disconnectedness is associated with severe health risks. Remarkably, the mortality risk of loneliness is similar to cigarette smoking, alcohol consumption, obesity, and physical inactivity (Holt-Lunstad and Steptoe, 2022). Thus, this highlights the importance of understanding the antecedents and implications of human-AI relationships, particularly for the sake of those populations where loneliness is most pronounced.

The current study examined Danish high-school students' sense of *social connectedness*, which covers a range of interrelated constructs regarding the quality of social relationships (Wickramaratne et al., 2022). It is commonly operationalized as low levels of loneliness and high levels of perceived social support (Wickramaratne et al., 2022). *Loneliness* denotes a distressing feeling of lacking intimate relationships (Hawkley and Cacioppo, 2010). According to the Loneliness Model, loneliness is driven by a hypervigilance for social threats, expectations of negative social interactions, and an inclination to remember information that reinforces the belief of being disconnected from others (Hawkley and Cacioppo, 2010). Loneliness is distinct from social isolation in that loneliness is only weakly to moderately associated with the actual frequency of social interactions and extensity of social networks (Taylor, 2020). In other words, socially isolated individuals are not necessarily lonely and vice versa, which highlights the role of cognitive

appraisals in the experience of loneliness.

A related concept is *perceived social support* that signifies the belief that others are available to provide care and appreciation (Singstad et al., 2021) or, as put elsewhere, the overall satisfaction with the available socioemotional support (Sarason et al., 1990). Perceived social support belongs to the umbrella concept of social support which encompasses a range of concepts describing the received or perceived availability of others (Haber et al., 2007). As with loneliness, perceived social support taps into the cognitive appraisal rather than the actual quality of social relationships, as it is only weakly associated with the received level of social support (Haber et al., 2007), yet more strongly related to well-being (Wills and Shinar, 2000). Despite this conceptual similarity, loneliness and perceived social support are only moderately correlated (Zhang and Dong, 2022), suggesting that each concept captures distinct nuances of social connectedness. Hence, to comprehensively assess youngsters overall sense of social connectedness, the current study operationalized it as loneliness and perceived social support.

2.2. Social compensation and displacement

In a digitalized world, one way to connect with others is online. In parallel with the growing popularity of social media and online communities within recent decades, a now well-established research field is investigating what the motivations are for connecting with others online, and what this means for people's social lives and well-being. An acknowledged proposition in this field—that might also explain the adoption of social chatbots—is *the social compensation hypothesis*. According to this hypothesis, humans turn to digital technologies to fulfill social needs when feeling socially disconnected, for example, due to isolation caused by social anxiety, shyness, or introversion (Ryan and Xenos, 2011; Toma, 2022). In other words, when experiencing that offline social network does not provide companionship, the social compensation hypothesis states that companionship is sought online. Accordingly, meta-analytical evidence shows a positive link between loneliness and Facebook usage and, notably, that loneliness positively predicts Facebook usage and not vice versa (Song et al., 2014). Research also indicates that loneliness predicts the number of online friendships through increased social networking and social compensation motives (Hood et al., 2018).

Social compensation behaviors might be attributed to the enhanced control of self-expressions during asynchronous interactions. Social media, internet forums, and gaming communities constitute a less intimidating environment than face-to-face interactions, as individuals do not become negatively influenced by physiological reactions related to anxiety, can take proper time to prepare their response, and shape their self-presentation in a more controlled manner (McKenna and Bargh, 2000; McKenna et al., 2002). All of this, coupled with the anonymity of many online platforms, offers a social arena that can be associated with less fear of judgment and rejection. In line with these ideas, research suggests that socially anxious individuals self-disclose more when communicating online than face-to-face as compared to non-socially anxious individuals (Weidman et al., 2012). Although online communities provide a more comfortable way to connect, highly socially anxious individuals who often communicate online experience lower life quality and more depression (Weidman et al., 2012), suggesting that online interactions may not fully compensate for deficits in face-to-face interactions.

Social compensation behaviors might also negatively influence well-being on its own. When using digital technologies to satisfy social needs, there is a risk of diminishing opportunities for face-to-face social interactions. The social displacement hypothesis posits that the time spent on digital devices replaces face-to-face social interactions with significant others (Hall and Liu, 2022). Social displacement is a concern, as research has shown that face-to-face interactions and relationships are more strongly linked to happiness and a sense of relatedness than online relationships and interactions (Helliwell and Huang, 2013; Vacchiano and Valente, 2021). Thus, social compensation and displacement may create a vicious cycle wherein individuals who struggle with social interactions become increasingly isolated when attempting to compensate through digital technologies. This highlights the pressing need to understand to what extent youngsters turn to chatbots to meet their social needs.

2.3. Human-AI relationships

While the social compensation hypothesis has been applied to explain why some individuals seek companionship online, recent research has shown evidence to suggest that this hypothesis also can be used to understand social chatbot adoption. Indeed, a new research field has emerged within recent years that inquires about the nature of human-AI relationships, including their antecedents and implications. By applying theoretical perspectives on interpersonal relationship development and characteristics, several qualitative studies have found that human-AI relationships resemble core features of interpersonal relationships, with regards to characteristics such as reciprocity and trust (Brandtzaeg et al., 2022; Skjuve et al., 2021). Additionally, a longitudinal interview study found evidence to suggest that human-AI relationships form largely consistent with the Social Penetration Theory (Skjuve et al., 2022), according to which human relationship formation is driven by increasing levels of mutual self-disclosures (Carpenter and Greene, 2015). Research has also highlighted potential antecedents of human-AI relationships. For example, Pentina et al. (2023) identified anthropomorphism (i.e., the tendency to attribute human traits to inanimate entities) and authenticity (i.e., the perception of the other as genuine, sincere, and honest) as significant predictors of perceiving interactions with Replika as inherently social and friendly. In turn, these perceptions were associated with the attachment felt toward Replika.

What remains largely uncertain, however, is whether users of social chatbots represent a group of individuals with particularly problematic social well-being. Seeking answers to this question is crucial, as it will influence our understanding of the societal and individual implications of social chatbots. For example, by facilitating hypothesis development concerning who is most at risk of suffering the consequences—or benefiting—from these technologies. To the best of our knowledge, only a few studies to date have examined social connectedness in social chatbot users broadly, while none have focused on youngsters specifically. Pentina et al. (2023) found that regular Replika users were significantly lonelier than college students in a cross-sample comparative analysis. Maples et al. (2024) also examined loneliness, though in college students using Replika, and found that the reported level of loneliness was higher than usually found in US college students in previous research. While these studies provide preliminary grounds for assuming that social chatbot users generally are lonelier than others, their findings must be interpreted with caution. What challenges these findings is the risk of sampling bias and selection bias resulting from different procedures and criteria to recruit social chatbot users and the comparisons. Some of these challenges can be addressed by conducting a large-scale observational study that identifies social chatbot users and investigates their overall sense of social connectedness compared to matching peers recruited through a similar procedure. Furthermore, there is a need to explore the role of social connectedness in social chatbot adoption to better understand why people use these technologies. As of yet, it remains to be understood whether the social compensation hypothesis applies to social chatbot adoption. The present study aims to address these gaps.

2.4. The present research

Considering the rising use of chatbots along with reports of high levels of loneliness in youngsters, the present research examined I) how many youngsters engage in social conversations with chatbots, ii) why

they engage in such interactions and iii) whether this usage relates to the sense of social connectedness. To pursue this aim, a preregistered cross-sectional survey study was carried out using a mixed-methods approach including a sample of Danish high-school students. The study entailed qualitative analyses to examine how youngsters socially engage with chatbots. Additionally, a series of confirmatory hypotheses were preregistered based on the social compensation hypothesis and research findings on the association between using Replika and reports of loneliness (Loveys et al., 2022; Maples et al., 2024; Pentina et al., 2023; Xie et al., 2023). First, we predicted that youngsters who converse with chatbots as friends are lonelier (H1a) and experience less perceived social support (H1b) than those who do not engage in friend-like conversations with chatbots. Second, we expected that higher loneliness (H2a) and lower perceived social support (H2b) positively predict the frequency of friend-like conversations with chatbots. Third, we preregistered hypotheses concerning why youngsters initiate conversations with chatbots. We hypothesized that loneliness predicts chatbot interactions intended to counteract loneliness (H3a), whereas low perceived social support predicts chatbot interactions intended to discuss one's feelings (H3b). Besides examining the confirmatory hypotheses, exploratory analyses were also carried out to explore the role of other potential triggers of chatbot usage, including boredom, bad mood, and a sense of friendship with the chatbot.

3. Method

3.1. Participants

All Danish high school students were invited to participate. To be eligible for inclusion, participants had to be enrolled in a Danish high school and be between 15 and 29 years old. In total, 14 educational institutions, with a total of 7.546 students enrolled, agreed to participate in the study. The educational institutions were geographically distributed across all major regions in Denmark, with four in the Capital Region, three in the North Jutland Region, three in the South Jutland Region, two in the Zealand Region, and two in the Central Denmark Region. A total of 2315 individuals accessed the survey, of which 1615 completed it, resulting in a response rate of 21.4 %. Eighteen responses were excluded due to falling outside the age criteria. Thus, 1599 responses were included for analysis. The mean age was 17.8 ($SD = 1.22$). Gender was evenly distributed, with 46.8 % male and 53.2 % female participants.

3.2. Procedure

The study was approved by the Institutional Review Board at Aarhus University (reference number: BSS-2023-129-S). Research ethical approval, preregistration, and supplementary materials are available on the Open Science Framework (https://osf.io/c938m/). The survey was delivered through the online survey platform RedCap. All Danish high schools registered on the official Danish homepage for educational guidance (https://www.ug.dk/) were invited to participate in the online survey with their students. The administrative units received an invitation letter via e-mail in early December 2023, which stated the purpose of the study and provided instructions on administering the survey to their students. In total, 235 educational institutions that offer high school programs were contacted. Non-responding institutions received one reminder after approximately 2 months. In compensation, the administrative units were informed that they would receive a written report detailing the overall study findings, as well as guidelines on how to discuss chatbot usage with their students. The students were not compensated for their participation. Although this decision might have reduced their willingness to participate, it was made to increase the reliability of the findings by minimizing the risk of low-effort responders participating solely for financial gain.

Data collection ran from December 2023 to March 2024. The survey link was administered to the students either during classroom lectures by the teachers or through closed online educational platforms for completion at home. To maximize the response rate, the educational institutions were encouraged to distribute the survey during classroom lectures while also allowing students the option to complete the survey at home. Although the differing settings for answering the survey could potentially influence students' responses systematically, we made this decision to increase the sample size, especially since some institutions were reluctant to participate if the survey had to be answered in class.

Informed consent was obtained from all participants. They were informed that the questionnaire included sensitive questions, that their responses were anonymous, their participation was voluntary, and they were free to withdraw at any time and without reason before submitting the response. Several precautions were taken to ensure participant anonymity. The survey did not collect internet protocol addresses. Participants were encouraged to refrain from describing any information that could identify them when answering open-ended questions. All free-text responses were read by the authors, and any identifiable information was pseudonymized. Once the participants had completed the survey, they were debriefed about its purpose and informed about available mental health resources.

3.3. Measurements

The survey consisted of a combination of standardized self-report tools and items specifically designed for this study. Pilot testing was conducted with four youngsters aged 15 to 18 to estimate if the length and content of the survey were fitting for the target group. The participants provided feedback on the newly developed items, and two of those items were slightly rephrased to enhance their clarity for this population.

3.3.1. The frequency of chatting with chatbots as friends

A single item was developed to assess the extent to which participants have conversed with chatbots as friends in the past month, using a 6-point Likert scale ranging from "Never" to "Every day". Specifically, participants were asked to rate the following item: "Have you within the past month chatted with a chatbot in the same way one would otherwise chat with a friend?".[1] Thus, we assessed the extent to which participants engage in friend-like interactions with chatbots, rather than explicitly asking participants if they would nominate a chatbot as a friend. This assessment strategy was selected to reduce the risk of social desirability bias influencing responses, which could occur due to the potential stigma of forming social relationships with chatbots (Skjuve et al., 2021). To aid comprehension of the scope of the item, it was explicitly stated that the item did not inquire about school-related usage of chatbots.

3.3.2. Triggers for initiating conversations with chatbots

Five items rated on 5-point Likert scales ("Strongly disagree" to "Strongly agree") were developed to assess psychological triggers for initiating conversations with chatbots. The assessed triggers were a) a feeling of loneliness ("I start a conversation with the chatbot when I feel lonely."); b) a need to self-disclose ("I start a conversation with the chatbot when I need to talk about how I'm feeling."; c) a feeling of boredom ("I start a conversation with the chatbot when I'm bored."; d) for emotional support when in bad mood ("I start a conversation with the chatbot because it supports me when I'm in a bad mood."); e) a feeling of friendship to the chatbot ("I start a conversation with the chatbot because I feel a friendship towards it."). The three latter triggers were examined in an exploratory manner. Participants also had the opportunity to provide a free-text description of other potential triggers for starting conversations with chatbots.

[1] Item quotes are translated ad verbatim from Danish.

3.3.3. Perceived social support

Perceived social support was assessed using the Danish version of the Multidimensional Scale of Perceived Social Support (Jakobsen et al., 2022; Zimet et al., 1988). It operationalizes perceived social support as a three-dimensional construct encompassing perceived social support from *family*, *friends*, and *significant others*. In total, the instrument consists of 12 items each rated on a 5-point Likert scale, ranging from ``very strongly disagree'' to ``very strongly agree''. The instrument has demonstrated good structural validity, construct validity, and internal reliability in Danish adult cardiac patients (Jakobsen et al., 2022) and American urban youngsters (Canty-Mitchell and Zimet, 2000). The full scale demonstrated high internal reliability ($\alpha = 0.936$).

3.3.4. Loneliness

Loneliness was assessed with the Danish version of the UCLA Loneliness Scale (Lasgaard, 2007; Russell, 1996). UCLA encompasses 20 items each rated on a 4-point Likert scale, ranging from ``Never'' to ``Often''. The Danish version of UCLA has shown satisfactory structural validity, construct validity, and internal reliability in Danish youngsters aged 14 (Lasgaard, 2007). The scale also demonstrated high internal reliability in this study ($\alpha = 0.914$).

3.4. Qualitative analysis

The qualitative analysis aimed to identify key conversation topics with chatbots. Besides providing qualitative insights into the purposes for which high-school students engage with chatbots as conversation partners, the qualitative findings were also utilized to statistically explore patterns of social connectedness across different ways of using chatbots. Analyses were conducted on participants' free-text answers to the following open-ended question: "Describe below what you usually chat with the chatbot about. Feel free to describe different topics you typically discuss with the chatbot, and as detailed as possible". Inspired by the guidelines outlined by Fereday and Muir-Cochrane (2006) and Braun and Clarke (2006), we qualitatively analyzed participants' responses using a hybrid inductive-deductive approach accompanied by statistical assessments of intercoder reliability.

The analytical process entailed five phases. In the first phase, the first author inductively analyzed participants' descriptions of how they engage with chatbots until thematic saturation was achieved. This phase aimed to construct a coding manual that provided an overview, definitions, and examples of the inductively derived codes. The second phase involved deductive coding of participants' descriptions using the coding manual, which was carried out independently by the first author and a research assistant at the master level. Participants' responses could be assigned multiple codes, for example, if describing more than one conversation topic. In the third phase, codes were hierarchically organized into higher-order codes based on their conceptual patterns to categorize students' conversations with chatbots. In the fourth phase, intercoder reliability was assessed for each independently assigned higher-order code. Disagreements were resolved by discussion. If there was less than a 'moderate' level of agreement for a higher-order code (i.e., *kappa* >0.60 (McHugh, 2012)) the coders engaged in collaborative reflection on the definition of the code, revised the coding manual accordingly, and repeated the deductive coding in conjunction with the revised definition. In the final phase, the first author consolidated the higher-order codes into categories representing core conceptual patterns of how high-school students engage with chatbots. By employing this hybrid inductive-deductive analytical approach, a data-driven analysis was ensured while also enhancing the reliability of the findings.

3.5. Quantitative analyses

Statistical analyses were conducted using IBM SPSS, version 28. Analyses were two-tailed with the significance level set at 0.05. A series of confirmatory and exploratory statistical analyses were carried out to examine the hypotheses. First, independent *t*-tests were conducted in line with the preregistration to examine whether chatbot users reported higher loneliness (H1a) and lower perceived social support (H1b) than their peers. The independent variable was operationalized as a dichotomous measure based on participants' self-reports of their chatbot interactions in the last month (see Section 3.3.1). Participants were classified as `chatbot users' if they reported engaging in conversations with a chatbot either ``once,'' ``a couple of times,'' ``every week,'' ``several times weekly,'' or ``daily.'' In contrast, 'non-users' were those who reported no chatbot interactions during the past month. To explore more fine-grained relationships between chatbot usage and social connectedness, we preregistered exploratory hierarchical regression analyses inserting qualitatively derived conversation topics as the independent variable. However, we deviated from the preregistered plan, as the qualitative analysis (see point 4.1) revealed that chatbot users engage in conversations on significantly heterogeneous topics. This contraindicated treating chatbot users as a homogeneous group and therefore compromised the validity of a dichotomous measure of chatbot usage. Hence, to enable comparisons within conversation topics and across students who engaged in social interactions with chatbots, we opted to conduct one-way Analyses of Variance (ANOVA). For this purpose, we computed a variable categorizing participants as non-users, social-supportive chatbot users, or utilitarian users based on the qualitative findings, and included loneliness and perceived social support as dependent variables in separate analyses. Tukey corrected post hoc tests were conducted to adjust for family-wise errors. Post hoc independent *t*-tests were performed to calculate standardized mean differences (*d*). Preliminary assessments were carried out to ensure compliance with the assumptions of one-way ANOVA tests, including normality and homogeneity.

Second, preregistered hierarchical regression analyses were conducted to examine the relationship between the frequency of chatbot usage and social connectedness (H2a; H2b) while controlling for age and gender. Separate analyses were performed for the two dependent variables: Perceived social support and loneliness. In Model 1, age and gender were inserted as control variables. In Model 2, a continuous measure of the frequency of chatbot interactions over the past month was inserted. Participants who reported no social interactions with a chatbot in the last month were excluded from this variable to mitigate skewness in the distribution of observations, as including non-users could introduce a disproportionate number of the lowest value and lead to a negatively skewed distribution. Preliminary assessments were conducted to ensure compliance to the assumptions of hierarchical regression analyses, including linearity, normality, independence of residuals, homoscedasticity, and non-collinearity.

Third, preregistered hierarchical regression analyses were performed to examine the relationship between social connectedness and various potential triggers of social conversations with chatbots, while controlling for age and gender. To test hypothesis H3a, age and gender were entered in Model 1, loneliness in Model 2, with loneliness as a trigger for chatbot conversations being used as the dependent variable. To test hypothesis H3b, age and gender were entered in Model 1, perceived social support in Model 2, and the need for self-disclosure as a trigger for chatbot conversations was used as the dependent variable. The same procedure was carried out for the exploratory analyses of boredom, friendship, and a bad mood as triggers of chatbot conversations. Furthermore, it is important to note that, due to the observed conceptual heterogeneity in chatbot conversation themes, all regression analyses were focused on youngsters who engage in social and/or supportive interactions with chatbots. This decision was guided by the social compensation hypothesis, which suggests that youngsters seeking social or emotional support from chatbots may do so to compensate for perceived deficits in their offline social interactions (Toma, 2022). This approach provides insights into the social connectedness of this specific group. However, the trade-off of heightened conceptual specificity is an increased sensitivity to measurement errors due to the relatively few

participants assigned to the group of social-supportive chatbot users (this issue is further discussed in the limitations section, see point 5.3). For transparency, hierarchical regression analyses that include the entire group of chatbot users are presented in the Appendix.

4. Results

Overall, the analyses showed that students who use chatbots are more socially disconnected than their peers. However, we also found that students who engage in casual conversations and seek emotional support from chatbots are even more socially disconnected than peers who do not use chatbots and more than those who use chatbots for utilitarian, goal-oriented purposes. The more socially disconnected students were more inclined to use chatbots upon situational feelings of loneliness, when in bad mood, and when feeling a need to self-disclose. See Table 1 for an overview of demographics and chatbot usage. We begin by summarizing the qualitative findings, as they inform the subsequent statistical analyses.

4.1. Conversation topics with chatbots

On average, participants' descriptions of conversation topics were 75.3 characters ($SD = 77.9$, *range* = 1–650). Seven subthemes emerged from the qualitative analysis. These subthemes were hierarchically organized into two overarching themes conceptualizing the conversation topics with chatbots: 1) Utilitarian conversations and 2) Social-supportive conversations. Furthermore, even though statistical analysis at conversation topic level could provide informative insights into fine-grained relationships with social connectedness, a dichotomous categorization was chosen to increase statistical power and decrease the sensitivity of group means to random individual variance. Of note, two responses were considered unserious and were therefore omitted from further analysis. Responses that were insufficiently informative to be interpreted (e.g., "wdjqwflka" and "all sorts of things") were assigned to the theme 'Miscellaneous' (Table 2).

Table 1
Demographics and chatbot usage.

	All	Chatbot users	Non-users
Number of participants[a]	1599 (100 %)	234[b] (14.6 %)	1365 (85.4 %)
Gender			
Male	748 (46.8 %)	125 (53.4 %)	623 (45.6 %)
Female	851 (53.2 %)	109 (46.6 %)	742 (54.4 %)
Age, *M* (*SD*)	17.8 (1.22)	17.9 (1.29)	17.8 (1.21)
≤16	182 (11.4 %)	18 (7.7 %)	164 (12 %)
17	497 (31.1 %)	83 (35.5 %)	414 (30.3 %)
18	539 (33.7 %)	70 (29.9 %)	469 (34.4 %)
19	312 (19.5 %)	51 (21.8 %)	261 (19.1 %)
≥20	69 (4.3 %)	12 (5.1 %)	57 (4.2 %)
Chatbot usage in past month, *M* (*SD*)	1.32 (0.90)	2.16 (1.22)	
Not at all	1365 (85.4 %)		
Once	75 (4.7 %)	75 (32.1 %)	
A couple of times	108 (6.8 %)	108 (46.2 %)	
Every week	10 (0.6 %)	10 (4.3 %)	
Several times weekly	20 (1.3 %)	20 (8.5 %)	
Every day	21 (1.3 %)	21 (9 %)	
Chatbot variants[c]			
ChatGPT	162 (10.2 %)	162 (69.2 %)	
Snapchat MyAI	94 (5.9 %)	94 (40.2 %)	
Character.ai	11 (0.7 %)	11 (4.7 %)	
Replika	3 (0.2 %)	3 (1.3 %)	
Other[d]	8 (0.5 %)	8 (3.4 %)	

Note.
[a] Unless otherwise stated, each row presents *numbers* and *percentages* of all participants for each column.
[b] In total, 248 participants stated they had chatted with chatbots as friends within the last month. However, fourteen were subsequently excluded from this category because they explicitly clarified in their free-text responses that they had not engaged with chatbots in a friend-like manner.
[c] Participants could select multiple answers and also had the opportunity to state in free-text responses which chatbots they use.
[d] The 'other' category includes chatbots with ≤2 users (i.e., Bing, Bard, Paradot, Copilot, and Pi.ai).

4.1.1. Social-supportive conversations

In total, 39 high-school students (16.7 % of chatbot-users) indicated that their conversations with chatbots have a social or supportive nature. Rather than the chatbot interactions serve a utilitarian or instrumental purpose, a shared feature among participants falling under this theme was that chatbots seemed to serve more social or emotional-supportive functions. We clarify this further below.

4.1.1.1. Social and personal problems. Some chatbot users described using chatbots to discuss various problems of a social or personal nature. Participants for example described intimate personal experiences (e.g., "my love life and about me being bullied"). Others stated that they discuss personal problems occurring in daily life and specifically seek out chatbots when feeling frustrated (e.g., "problems, personal issues, daily problems, when I am frustrated about something, when I have a question". Participants also stated that they not only use chatbots as a means to cope with negative emotions but also for self-improvement (e.g., "divorce, self-esteem, discipline, good habits, and mental health"). However, the latter participant also stated, "it's really bad, and I've started to stop writing because it doesn't matter much", indicating that the sustained adoption of chatbots to discuss social and personal problems is challenged by problematic user experiences in some individuals.

4.1.1.2. Advice. Participants assigned to this subtheme sought advice from chatbots on various topics, some of which did not specify what they needed advice for. Some described how they use chatbots to help them interpret situations and get advice for how the situations can be dealt with (e.g., " [I] give situations from my life and what they can mean, and how I should handle them" and "about advice on what I should do in situations"). Others framed the chatbots as assistants that help them make decisions in their everyday lives that one could otherwise discuss with friends or family (e.g., "if I find it difficult to make a decision, for example, about where to travel, what to do for dinner, etc.").

4.1.1.3. Casual conversations. Some chatbot users described having casual conversations with chatbots, for example small-talk or intellectual conversation topics such as politics or history. Responses assigned to this theme distinguished themselves by indicating that the chatbot interactions were of a reciprocal nature. In other words, instead of merely using the chatbot's *output* for a specific goal, the purpose of the interaction appeared to be mutual participation in a conversation. One participant, for example, wrote: "I usually chat with the chatbot about topics that interest me, such as history. The reason for this could be, for

Table 2
Overview of conversation topics with chatbots.

Categories	Topics	N[a]	%	Kappa
Social-supportive use	Social and personal problems	16	6.8 %	0.802***
	Advice	10	4.3 %	0.817***
	Casual conversations	14	6 %	0.799***
Utilitarian use	School-related activities	87	37.2 %	0.958***
	Information-searching	70	29.9 %	0.802***
	Entertainment	47	20.1 %	0.882***
	Generating ideas	16	6.8 %	0.648***
Miscellaneous		19	8.1 %	0.892***

Note.
[a] Participants could be assigned to more than one subtheme.
*** $p < 0.001$.

example, that I don't know many people who are as interested in history as I am.". What became apparent here is that the student utilizes the chatbot to compensate for a lack of friends who share their interests. Other participants described a variety of casual conversations topics reflecting a tendency to small-talk with chatbots, e.g. "everyday things, such as the weather, hobbies, etc.", "e.g., How has your day been?", and "all sorts of different things, politics is always a good topic of conversation".

4.1.2. Utilitarian conversations

In total, 174 high-school students (76.5 % of chatbot users) described more utilitarian/instrumental ways of interacting with chatbots. Responses falling under this theme illustrated practical, goal-oriented purposes of engaging with chatbots, hereunder completing school-related tasks, searching for information, entertainment, and generating ideas. For these students, chatbots apparently serve as a new online tool that instantly can provide detailed responses to an almost unlimited variety of questions. As such, the majority of participants seemed to describe their use of chatbots more as *tools* than as *relational entities*. We outline and exemplify each subtheme below.

4.1.2.1. School-related conversations. The majority of chatbot users described school-related conversations. Participants for example wrote "schoolwork", "often for summarizing texts, or corrections regarding grammar etc. in my assignments", and "when I need help with an assignment or when I need an idea for an assignment". The purpose of the interactions falling under this subtheme was to solve tangible tasks related to schoolwork, and there were no indications that the interactions moved beyond a merely utilitarian function.

4.1.2.2. Information-searching. The second most frequent topic was information searching. Responses assigned to this subtheme (e.g., "information search") used chatbots to obtain information for a variety of purposes that were not stated to serve socio-emotional supportive functions or help complete school-related tasks. Participants described how they use chatbots as an alternative to search engines such as Google (e.g. "normal searches just like when one uses Google"). Others highlighted the speed of chatbots' responses, noting that they could "ask questions about things and get a quick answer."

4.1.2.3. Entertainment. A considerable number of chatbot users described using chatbots as toys for entertainment. Rather than explicitly stating they had fun *together* with the chatbot, these participants indicated that they use the chatbot's *outputs* for entertainment. For instance, participants wrote "… asks intentionally stupid questions to see how it reacts". Other highlighted how chatbot outputs simply function to cope with boredom (e.g., "just trying to get it to say some stupid things when I'm bored". Finally, some described how they use chatbots for entertainment by prompting them to assume different roles (e.g., "I got character.ai to be a cartoon character for fun, there's not that much to it").

4.1.2.4. Idea-generation. Some participants described that they use chatbots' outputs as inspiration for a variety of tasks, for example, "as an idea generator for different perspectives on arguments within, for example, IT projects". Others described how chatbots can help them get inspiration for handling interpersonal interactions (e.g., "I get it to write a good rejection LMAOoo"). Participants also explained how chatbots serve as tools for getting ideas for hobbies (e.g., "I often use ChatGPT for Dungeon and Dragons stuff to help with that and come up with ideas and code").

4.2. Social connectedness among chatbot users

4.2.1. Comparative analyses

We observed significant differences in social connectedness depending on *whether* and *how* the students engage with chatbots (See Table 3 for an overview of pairwise comparisons). This subsection initially presents the findings of our preregistered comparisons of chatbot users and non-users on social connectedness. Subsequently, we display the results of exploratory analyses comparing diverse ways of using chatbots on social connectedness.

Overall, analyses showed that chatbot users are more socially disconnected than their peers. Chatbot users were significantly lonelier (H1a) and experienced less social support (H1b) than non-users. However, while chatbot users (M = 41.0, SD = 10.1) were significantly lonelier than non-users (M = 39.5, SD = 9.79, $t(1597) = 2.08$), the effect size was insufficient to be considered 'small' according to Cohen's rule of thumb[2] (See Table 3). Chatbot users (M = 62.5, SD = 13.1) also experienced significantly less social support than non-users (M = 64.6, SD = 12.8, $t(1597) = -2.38$), yet the effect size neither reached the threshold for a small effect. As such, although both preregistered hypotheses were supported, the findings only suggest minor differences in social connectedness between users and non-users of chatbots.

However, as also described in Section 3.5, the heterogeneity of conversation topics with chatbots contraindicated treating chatbot users as one homogeneous group. To potentially delineate more fine-grained associations between chatbot usage and social connectedness, non-preregistered one-way ANOVA tests were carried out comparing 1) social-supportive chatbot use, 2) utilitarian chatbot use, and 3) non-users of chatbots on measures of social connectedness as dependent variables. See Fig. 1 for a bar chart illustrating the differences in social connectedness across the three groups.

We identified a significant omnibus effect on loneliness across the three groups: social-supportive use (n = 39, M = 44.7, SD = 11.6), utilitarian use (n = 174, M = 39.6, SD = 9.32), and non-users (M = 39.5, SD = 9.79, $F(2, 1575) = 5.38$, $p = .005$, $n^2 = 0.007$).[3] Post hoc tests with Tukey corrections, controlling for family-wise error rates, showed that social-supportive users were significantly and moderately lonelier than non-users (see Table 3). They were also significantly and moderately lonelier than utilitarian users. Loneliness was not significantly different between utilitarian users and non-users. The analyses highlight that a dichotomous categorization failed to capture fine-grained nuances in

Table 3
Pairwise comparisons.

	Loneliness			Perceived social support		
Comparisons	p	d[a]	95 % CI	p	d[a]	95 % CI
Chatbot users vs. non-users	0.037	0.15	0.09, 0.29	0.018	−0.17	−0.31, −0.03
Social-supportive vs. non-users	0.003	0.53	0.21, 0.85	0.012	−0.46	−0.78, −0.14
Social-supportive vs. utilitarian	0.009	0.52	0.17, 0.87	0.074	−0.38	0.03, 0.73
Utilitarian vs. non-users	0.992	0.01	−0.15, 0.17	0.620	−0.08	−0.23, 0.83

Note.
[a] Effect sizes were calculated by running independent t-tests.

[2] According to Cohen's (1988) rule of thumbs, standardized mean differences (d) can be interpreted as follows: small: $d \geq 0.20$; moderate: $d \geq 0.50$; large: $d \geq 0.80$.

[3] Beyond the reduced statistical power leading to an increased rate of Type II errors, unequal sample sizes can also cause unequal distributions of residuals. Therefore, Levene's tests were conducted for all comparisons, and the results did not indicate significant differences in residual variance across the groups.

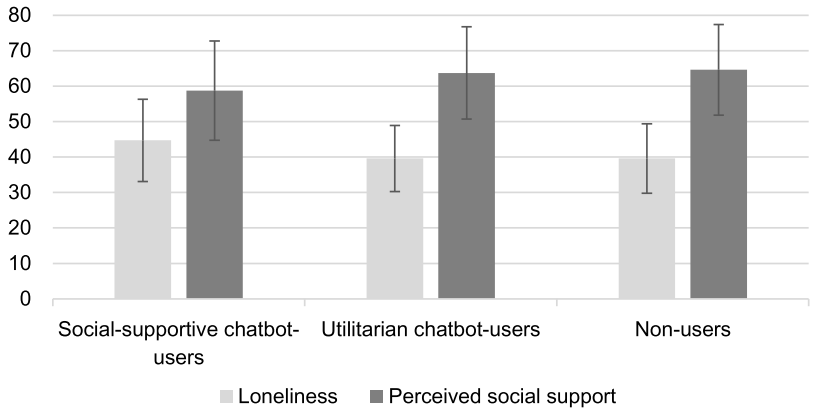


Fig. 1. Social connectedness across whether and how participants use chatbots.

loneliness, as expressed in the distinct levels of loneliness between utilitarian and social-supportive chatbot users.

A similar statistical pattern was observed in perceived social support. A significant omnibus effect on perceived social support was observed across social-supportive users ($M = 58.7$, $SD = 14.0$), utilitarian users ($M = 63.7$, $SD = 13.0$), and non-users of chatbots ($M = 64.6$, $SD = 12.8$, $F(2, 1575) = 4.35$, $p = .013$, $n^2 = 0.005$). Post hoc tests with Tukey corrections controlling for Family-wise errors rates showed that social-supportive chatbot users reported significantly less perceived social support than non-users, with the analysis revealing a small effect size (see Table 3). Social-supportive users reported significantly less perceived social support than non-users of chatbots. The analysis revealed a small effect size, suggesting a somewhat minor difference in perceived social support. Moreover, a small though insignificant difference was observed between social-supportive and utilitarian chatbot users, suggesting no reliable difference. Finally, no significant difference was observed between utilitarian users and non-users. In sum, the findings suggest that students who engage with chatbots in a friend-like fashion do not per se feel more socially disconnected than their peers. Rather, students who engage in *social-supportive conversations* with chatbots are those who reported feeling more socially disconnected.

4.2.2. The relationship between the chat frequency and social connectedness

Overall, we did not find evidence to suggest that students who more frequently interact with chatbots are also more socially disconnected. See Table 4 for zero-order correlations between measured variables among the social-supportive chatbot users. As argued previously, the analyses were carried out on the social-supportive chatbot users (see Appendix for results for the full group of chatbot users).

First, we present the results of a hierarchical regression analysis of the relationship between *loneliness* and *chat frequency* while keeping age and gender as covariates (H2a; see Table 5 for an overview of results). Model 1, including age and gender, did not significantly predict chat frequency ($F(2, 36) = 3.12$, $R^2 = 0.148$, $p = 0.057$). While age was insignificant ($\beta = 0.09$, $t = 0.60$, $p = 0.953$), gender was a significant predictor ($\beta = 0.39$, $t = 2.49$, $p = 0.017$). Model 2 (including age, gender, *and* loneliness) neither predicted chat frequency significantly ($F(3, 35) = 2.02$, $p = 0.129$). In model 2, gender remained a significant predictor ($\beta = 0.38$, $t = 2.42$, $p = 0.021$), while age remained insignificant ($\beta = 0.09$, $t = 0.06$, $p = 0.953$). Loneliness was neither a significant predictor of chat frequency ($t = 0.02$, $p = 0.984$). Likewise, model 2 did not demonstrate a significant improvement over model 1 ($\Delta p = 0.984$).

Second, we outline the results of a hierarchical regression analysis of the relationship between *perceived social support* and *chat frequency* (H2b). Model 1 (including age and gender) did not significantly predict chat frequency. Model 2 (including age, gender, and perceived social support) neither predicted chat frequency significantly ($F(3, 35) = 2.12$, $p = 0.115$). Again, gender remained a significant predictor ($\beta = 0.41$, $t = 2.51$, $p = 0.017$), whereas age was statistically insignificant ($\beta = 0.02$, $t = 0.10$, $p = 0.923$). Perceived social support was not a significant predictor of chat frequency ($t = 0.52$, $p = 0.608$). As displayed in Table 5, model 2 did not demonstrate a significant improvement over model 1 ($\Delta p = 0.608$).

To summarize, the results conflicted with the idea that loneliness and perceived social support are associated with how often Danish high-school students engage in conversations with chatbots. As can be seen in the Appendix, the same pattern was observed in the full group of

Table 4
Descriptive statistics and zero-order correlations for social-supportive chatbot users ($n = 39$).

	M	SD	1	2	3	4	5	6	7	8	9
1. Age[a]	17.6	0.99									
2. Gender	1.56	0.50	−0.059								
3. Chat frequency	2.03	1.29	−0.013	0.384*							
4. Loneliness	44.7	11.6	−0.021	0.177	0.071						
5. Social support	58.7	14.0	−0.051	−0.314	−0.045	−0.614***					
6. T[b]: Loneliness	1.87	1.24	−0.060	−0.008	0.316	0.390*	−0.232				
7. T: Boredom	3.26	1.29	.095	−0.026	0.059	0.038	−0.300	0.136			
8. T: Disclosure	2.51	1.39	0.406*	−0.161	0.110	0.202	−0.333*	0.283	0.320*		
9. T: Friendship	1.51	0.756	−0.135	0.119	0.229	0.205	−0.168	0.549***	0.212	0.268	
10. T: Bad mood	1.92	1.16	0.114	0.167	0.161	0.517***	−0.448**	0.507***	0.102	0.581***	0.498**

Note.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.
[a] Gender: 1 = female, 2 = male.
[b] T: Trigger.

Table 5
Hierarchical regression analyses of chat frequency and triggers among social-supportive chatbot users ($n = 39$).

	Loneliness				Perceived social support			
Dependent variable	R^2	ΔR^2	ΔF	β	R^2	ΔR^2	ΔF	β
Chat frequency	0.148	0.000	0.00	0.03	0.154	0.006	0.27	0.09
Trigger: Loneliness	0.161	0.158	6.58*	0.40*	0.067	0.063	2.37	−0.27
Trigger: Disclosure	0.240	0.057	2.63	0.24	0.325	0.141	7.31*	−0.40*
Trigger: Boredom	0.011	0.002	0.07	0.05	0.111	0.102	4.01	−0.34
Trigger: Friendship	0.065	0.035	1.29	0.19	0.052	0.022	0.80	−0.16
Trigger: Bad mood	0.290	0.247	12.2**	0.51**	0.210	0.142	7.38*	−0.43*

Note.
* $p < 0.05$.
** $p < 0.01$.
***$p < 0.001$. The table displays results from model 2 of each hierarchical regression analysis.

chatbot users. Thus, the findings were inconsistent with hypotheses 2a and 2b.

4.2.3. Triggers of chatbot conversations

The final objective of this research was to examine the relationship between social connectedness and triggers of engaging in conversations with chatbots. A series of preregistered confirmatory and exploratory hierarchical regression analyses were carried out. The confirmatory analyses are outlined below and in Table 5. The results of the exploratory analyses are only displayed in Table 5.

First, we present the analysis of the relationship between overall loneliness and a sense of loneliness as a trigger for engaging in conversations with chatbots (H3a). Model 1 (including age and gender as covariates) was not statistically significant ($F(2, 36) = 0.07$, $R^2 = 0.004$, $p = 0.935$). Neither age ($\beta = -0.06$, $t = -0.37$, $p = 0.717$) nor gender ($\beta = -0.01$, $t = -0.07$, $p = 0.947$) were significant predictors. Model 2 (including age, gender, and loneliness) was also statistically insignificant ($F(3, 35) = 2.24$, $R^2 = 0.161$, $p = 0.008$). Yet, loneliness was a significant predictor ($t = 2.56$, $p = 0.015$), whereas age ($\beta = -0.06$, $t = -0.37$, $p = 0.717$) and gender ($\beta = -0.08$, $t = -0.52$, $p = 0.605$) remained insignificant. Furthermore, model 2 yielded a significant improvement from model 1 ($\Delta p = 0.015$), explaining 15.8 % additional variance in the dependent variable.

Second, we present the analysis of the association between perceived social support and the need for self-disclosure as a trigger of chatbot use (H3b). Model 1 (including age and gender as covariates) was statistically significant ($F(2, 36) = 4.04$, $R^2 = 0.18$, $p = 0.026$. While age was a significant predictor ($\beta = 0.40$, $t = 2.64$, $p = 0.012$), gender was not ($\beta = -0.14$, $t = -0.91$, $p = 0.367$). Model 2 (including age, gender, and perceived social support) was also found to be significant ($F(3, 35) = 5.61$, $p = 0.003$). While gender remained insignificant ($\beta = -0.26$, $t = -1.80$, $p = 0.081$), age was a significant predictor ($\beta = 0.37$, $t = -2.65$, $p = 0.012$). Notably, perceived social support was also statistically significant ($t = -2.70$, $p = 0.010$). Adding perceived social support to model 2 yielded a significant improvement over model 1 ($\Delta p = 0.010$) and uniquely accounted for 14.1 % of the variance in the dependent variable.

In other words, the findings suggest that individuals with higher levels of loneliness are more inclined to seek out chatbots when experiencing a situational feeling of loneliness. Likewise, students experiencing less social support are more inclined to start conversations with chatbots when feeling a need for self-disclosure. A similar pattern was observed in the total group of chatbot users (see Appendix). Therefore, the results were consistent with hypotheses 3a and 3b.

5. Discussion

Recent years have featured the launch of social chatbots capable of engaging in human-sounding conversations. We may therefore face a future where social chatbots (as well as social robots and digital avatars) may increasingly enter our social lives. This has sparked concerns that chatbots may particularly attract lonely individuals seeking companionship and support. To the best of our knowledge, the present preregistered research represents the first scientific effort to systematically explore the adoption of chatbots as conversation partners in youngsters specifically, and more broadly, humans in general. Utilizing a mixed-methods approach, this cross-sectional study searched for preliminary answers to how many and why youngsters engage in social conversations with chatbots, and how this trend relates to the sense of social connectedness.

5.1. The adoption of social chatbots in youngsters

A core contribution of this study is an estimate of how many youngsters currently engage in social interactions with chatbots. While the launch of LLM chatbots such as ChatGPT, Replika, and MyAI has sparked concerns regarding the consequences of chatbot use, to the best of our knowledge, no estimates of the adoption of social chatbots existed within such a population. Without these estimates, it is uncertain to what extent there are grounds for concerns, such as the risk for users to become dependent on chatbots and displacing face-to-face interactions. However, unraveling the adoption of social chatbots proved more challenging than expected. In broad terms, these challenges reflect the complexities of demarcating what social interactions are. While 14.6 % of students reported having friend-like conversations with chatbots during the past month, the qualitative analysis showed that 10.9 % reported using chatbots for school-related activities, information-searching, generating ideas, or as toys for entertainment. On the other hand, only 2.44 % of the total sample indicated that their conversations were of a social or supportive nature. Assuming that our sample represents the total population of Danish high-school students ($N = 141.511$; Statistics Denmark, 2023), the latter figure equals 3.453 students. However, these estimates should be interpreted with caution due to the potential for self-selection bias among youngsters with a particular interest in social chatbots, given the low response rate of 21.4 %.

Despite this possible bias, a striking observation was that the majority of chatbot users reported on using chatbots as utilitarian tools rather than relational entities, even though the study explicitly inquired about friend-like conversations with chatbots. These findings could be interpreted as some participants having misunderstood the scope of the survey. Perhaps they believed the survey inquired about *general* chatbot usage, maybe reflecting that other uses of chatbots than utilitarian ones fall entirely outside their frame of reference. However, it could not be ruled out that their responses indeed reflected diverse understandings of what social interactions are, for example that they discuss school-related tasks in a conversation style alike to conversations with peers.

A similar issue is that it remains quite unclear to what extent social-supportive chatbot users engage with chatbots as relational entities rather than as tools for emotional support. Only 0.88 % of the total sample indicated that their conversations with chatbots are characterized by mutual participation in conversations. On the other hand. 1.63 % explained that the conversations revolve around social and personal

problems and advice for dealing with issues. This finding is particularly challenging to interpret, as it cannot be concluded whether the function of chatbots for these students resembles that of empathetic relational partners. Indeed, a conflicting idea is that the chatbots serve as tools similar to informational webpages about mental and social issues, with chatbots primarily distinguishing themselves by providing more personalized information. This idea contests the validity of the social compensation hypothesis in a human-chatbot context, as it implies that the motivation for turning to chatbots is not necessarily to fulfill unmet social needs but to seek information on how to deal with various emotional and social issues. While some Danish youngsters seem to engage in mutual participative conversations with chatbots, this trend appears to be relatively minimal.

However, the generalizability of Danish high school students to other demographics, nationalities, and cultures should be called into question. In particular, several factors may influence the adoption of new technologies such as access to technology, digital literacy (Nikou et al., 2022), and technology readiness (Sunny et al., 2019), as well as factors related specifically to the needs that social chatbots may address, such as the levels of social disconnectedness, isolation, and alienation. Denmark has a highly developed digital infrastructure (European Comission, 2022), a degree of technology adoption, digital literacy, and digital readiness that equals or exceeds other Western countries (Mancino, 2023; World Bank, 2016), as well as a high adoption of social media in youngsters (Kulturministeriet, 2021). At the same time, the degree of loneliness in youngsters is, as mentioned, at a similar level as other Western countries (Lasgaard et al., 2016; Office for National Statistics, 2018), which according to the social compensation hypothesis would predict an increased inclination to use social chatbots. While such insights give the impression that Danish high-school students reliably generalize to other Western youngsters, there might be important yet unknown factors beyond digital literacy and loneliness that influence social chatbot adoption cross-culturally. For example, animistic attributions (i.e., ascribing life to inanimate entities (Karpinska-Krakowiak and Eisend, 2021) have been found to be significantly more prevalent in individuals with East Asian than European decent (Folk et al., 2023). Another potential mechanism is the tendency to feel threatened by unknown or ambiguous situations, known as uncertainty avoidance, where artificial entities are anthropomorphized to achieve a sense of explainability and predictability (Epley et al., 2007). As with animistic attributions, uncertainty avoidance has been shown to differ significantly between East Asian and Western Countries, with the latter generally scoring lower in this trait (Yi, 2021). Hence, while Western populations may generally share low levels of animistic thinking and uncertainty avoidance, these cultural traits might represent boundary conditions limiting the generalizability of our findings to other cultures. Thus, in the absence of more research within and between cultures, the estimates of social chatbot adoption should be generalized with caution.

5.2. Social connectedness in chatbot-users

The present research contributes to the understanding of the role of social connectedness and compensation for social chatbot adoption. We expand on earlier research showing indications that social chatbot users experience higher levels of loneliness than others (i.e., Maples et al., 2024; Pentina et al., 2023). For the first time, the present study provides evidence substantiating the presumption that youngsters who engage in social interactions with chatbots are more socially disconnected than their peers based on a large-scale cross-sectional analysis of a broad population sample. Furthermore, this study contributes to the field by identifying how different modes of interacting with chatbots are linked to varying levels of social connectedness. Indeed, students who specifically engage in casual conversations and discuss personal issues with chatbots generally report higher levels of loneliness and lower levels of perceived social support. This observation was noted not only in comparison to peers who do not use chatbots but also to those who use them for utilitarian purposes, even if they state that they converse with chatbots in a friend-like manner. In a similar vein, previous research has also shown that the purpose of using social chatbots varies, with some individuals using Replika merely as entertainment to combat boredom (Pentina et al., 2023). Hence, with the reservation that some of our participants might have misinterpreted the survey's scope, it seems that *whether* someone engage in friend-like conversations with chatbots or uses Replika may not reliably predict their social connectedness.

Another key contribution of the present study is its insights into the applicability of the social compensation hypothesis to social chatbot adoption. While the observed social connectedness among chatbot users does not firmly support the social compensation hypothesis due to difficulties in distinguishing cause from effect, the findings regarding the triggers of social chatbot interaction align with this hypothesis. Indeed, we found that lonelier students were more inclined to start conversations with chatbots specifically when experiencing situational loneliness. Moreover, students who perceive less social support were more inclined to use chatbots when feeling a need to self-disclose. Put differently, when individuals with lower levels of social connectedness experience an unfulfilled social or emotional need, they seem to start conversations with chatbots, as the social compensation hypothesis would predict. This finding aligns and expands on earlier qualitative findings showing that some individuals specifically use Replika to satisfy unfulfilled socioemotional needs, but also as a safe haven to discuss sensitive personal information without feeling stigmatized (Skjuve et al., 2022). Hence, both qualitative and quantitative findings collectively point to the idea that social compensation motivation promotes social chatbot usage.

What challenges the social compensation hypothesis in a human-AI context is the observation that more socially disconnected students do not seem to initiate conversations with chatbots because they perceive a sense of friendship. This finding holds implications for our understanding of the *role* that youngsters higher in social disconnectedness attribute to chatbots. What stands out from the observation that these students engage with chatbots when feeling lonely, sense a need for self-disclosure, or are in a bad mood—but not due to a sense of friendship—is that chatbots seem to serve instrumental functions as *coping tools* rather than as social companions. In broad terms, coping has an instrumental nature in that it encompasses cognitive and behavioral efforts to manage internal and external stressors (Folkman and Moskowitz, 2004). The efforts are only applied to solve a problem, and the efforts may cease once the problem is no longer there. A similar instrumentality is apparent in our findings. Rather than positive sentiments driving the chatbot interactions, relieving negative emotional states seems to do so. In a Kantian sense (Kant et al., 2002), chatbots may serve as a *means to an end* (i.e., coping with negative feelings) rather than the friendship itself being the end that is pursued. A theoretical implication of this possibility is that when the need to cope with negative feelings ceases, it might be expected that youngsters might discontinue reaching out to chatbots.

However, given the cross-sectional, observational nature of the present study, the findings do not contest the possibility that intimate human-AI relationships may evolve over time. As mentioned, previous studies indicate that such relationships may emerge at least in some individuals (Brandtzaeg et al., 2022; Skjuve et al., 2021), perhaps especially with prolonged use (Skjuve et al., 2022). In that sense, the motivation for engaging with chatbots may evolve from *avoidance motivation*, signifying the behaviors driven by a desire to avoid negative feelings, to *approach motivation* (Elliot, 2006), wherein the social interaction with chatbots becomes intrinsically motivating as the attachment strengthens. Our findings neither align with nor challenge such findings, as the participants in the present study generally had engaged with chatbots to a small degree. Among the social-supportive chatbot-users, 82.1 % (n = 32/39) had only engaged in conversations with chatbots once or a couple of times during the last month, indicating sporadic or discontinued use of chatbots as social or supportive agents. Negative

user experiences might explain this observation, as illustrated by one participant: "Asked it for advice, but they are not very well developed. So I quickly put it away from me again". Likewise, and contrary to our expectations, we observed no relationship between social connectedness and how often students chat with chatbots, which might reflect discontinued use due to negative user experiences, although the initial interaction perhaps was driven by socioemotional needs.

5.3. Limitations

There are several reasons why our findings should be interpreted with caution. The first limitation concerns the cross-sectional nature of this research. While a cross-sectional methodology offered increased ecological validity due to its observational nature and enabled a large-scale, geographically representative investigation, it limits the ability to draw firm causal conclusions. This limitation particularly applies to interpreting the observed differences in social connectedness based on whether and how the respondents engage with chatbots. The reason why Danish high-school students who use chatbots for social-supportive purposes are more socially disconnected than others therefore remains tentative. Efforts were made to address this limitation by assessing the triggers of chatbot interactions at the phenomenological level, which showed meaningful links to social connectedness. Nonetheless, longitudinal and experimental research is essential to further substantiate the suggested causal pathway between social disconnectedness and the triggers for chatbot interactions.

Second, the conceptual limitations of the group of 'social-supportive chatbot users' must be stressed. This group encompassed a diverse range of conversation topics that may be too heterogeneous to be considered members of a uniform conceptual category. Specifically, it included casual conversations, advice seeking, and discussing personal and social problems, signifying quite different ways of engaging with chatbots, conceptually speaking. As a consequence, the results associated with this group lack specificity for each type of conversational topic. This categorization approach was driven by our aim to create meaningful themes whilst also creating groups large enough for reliable statistical analysis, as previously mentioned. Analyzing all conversation topics separately would have compromised statistical power and increased the sensitivity of groups means to random individual variance. Nonetheless, the results demonstrated that our classification was meaningful. Specifically, participants whose conversations were grouped under the broad category of 'social-supportive chatbot use' distinguished themselves, albeit at the group level, from both utilitarian users and non-users in terms of social connectedness. However, more fine-grained patterns between social connectedness and specific conversation topics with chatbots remain unknown and merit further investigation.

Third, and related to the latter limitation, it cannot be ruled out that this research suffers from the so-called *winner's curse*. That is, the phenomenon specific to low-power studies in which an observed effect size is inflated relative to the 'true' effect size, for example, due to increased sensitivity of statistical parameters to random variance (Gelman et al., 2020). The group of social-supportive chatbot users was relatively small ($n = 39$), which raises concerns about the internal validity of the findings due to potential measurement errors. External validity may also be problematic, as the small group size limits the generalizability of the results to the broader population. In this particular case, this limitation may arise from sampling bias where the sample is not fully representative of the population in question, potentially due to factors such as self-selection bias and demographic imbalances within the group. The small group size makes the study more sensitive to such biases, so the findings should be considered preliminary and warrant further investigation.

The final limitation concerns the validity of our assessment of 'friend-like' conversations with chatbots, which is a matter that can and should be called into question. To acknowledge the diverse understandings of the characteristics of friendship (Kitts and Leal, 2021; Mendelson and Aboud, 1999; Nicolaisen and Thorsen, 2017), our assessment of 'friend-like conversations' relied on participants' subjective understandings of this concept. However, this resulted in a heterogeneous group, encompassing both social-supportive and utilitarian ways of interacting with chatbots. We addressed this issue by focusing on *how* rather than *whether* participants engage with chatbots in the statistical analyses and identified meaningful links to social connectedness. Nonetheless, advancing this research field requires the development of standardized measurement tools that meet the demands of *sensitivity* (i.e., capable of identifying individuals who engage in social conversations with chatbots) and *specificity* (i.e., able to avoid identifying individuals who do not engage in such conversations, for example, those using chatbots similarly to Google searches). To pursue this goal, conceptualizing social interactions in a way that allows for sound categorical operationalizations is a pressing issue. However, heterogeneous perceptions of what characterizes social interactions and relationships (Kitts and Leal, 2021; Policarpo, 2015) may complicate this endeavor. Another method to measure social interactions and relationships with chatbots could involve assessing whether individuals would *nominate* a chatbot as, for example, a friend. As mentioned earlier, we refrained from this approach to avoid the influence of social desirability bias on responses due to the stigma associated with befriending inanimate entities (Skjuve et al., 2021). Additionally, a nomination approach could be problematic in this particular context because, for some individuals, chatbots may not qualify for the label "friend" due to their ontological status as inanimate entities. Of note, this possibility also challenges the validity of our assessment of friendship as a trigger for chatbot interactions. The pressing need to develop measurement tools is further highlighted by the possibility of a near future in which chatbots, social robots, and virtual avatars become increasingly sophisticated and human-like. Valid and reliable measurement tools are of utmost importance to continually monitor the extent to which humans engage in social relationships with conversational technologies, along with the antecedents and implications of such relationships.

6. Conclusion and future directions

The present study was carried out to explore how many youngsters engage in social conversations with chatbots and their motivations for doing so. By pursuing these aims, the present study contributes preliminary insights into the adoption and socioemotional antecedents of social chatbot use among youngsters. In broad terms, our findings indicate that 2.44 % of high school students have had casual conversations or sought emotional support from chatbots in the past month, and, importantly, that these students are more socially disconnected than their peers. Hence, this research highlights a potential dark side of novel chatbots that warrants further investigation: The possibility that some socially disconnected youngsters turn to chatbots to cope with unmet social and emotional needs. In that sense, we propose that social chatbots are not necessarily utilized as *social companions* but rather as *tools* to handle internal and external stressors. This idea has broader implications for our understanding of the phenomenological status of social chatbots and similar technologies as agents or tools. While this study does not reject the notion that social chatbots can function as social companions, it suggests that, for some youngsters, chatbots may primarily serve as tools to cope with socioemotional deficits.

As particularly youngsters with higher levels of social disconnectedness seem to use chatbots for socioemotional purposes, there is a need for policymakers and industrial stakeholders to begin reflecting on the ethically responsible design and implementation of chatbots. In particular, this is important to help prevent social displacement caused by social chatbots. However, it is also necessary to seek answers to lingering epistemic questions to inform the development of such ethical guidelines. As of now, there is scarce evidence to anticipate the implications of social chatbots. In broad terms, future research should further explore the antecedents, moderators, and importantly, implications of seeking

social and emotional support in conversational technologies. In particular, there is a need for longitudinal experimental studies exploring how people engage with social chatbots over time, how their relationships evolve, and what this means for their sense of social connectedness. Furthermore, there is a need for methodological contributions aimed at exploring how to reliably assess social interactions with chatbots and similar conversational agents. The present study also calls for cross-cultural studies involving different demographic groups as cultural mechanisms might influence social chatbot adoption. All of these are crucial next steps to achieve a better understanding of whether—or under which conditions—social conversations with chatbots are helpful or harmful.

CRediT authorship contribution statement

Arthur Bran Herbener: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Malene Flensborg Damholdt:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ijhcs.2024.103409.

Appendix

Table A1.

Table A1
Preregistered hierarchical regression analyses of all chatbot users ($n = 234$).

	Loneliness					Perceived social support				
Dependent variable	R^2	ΔR^2	ΔF	Δp	β	R^2	ΔR^2	ΔF	Δp	β
Chat frequency	0.025	0.000	0.09	0.767	0.02	0.025	0.000	0.02	0.969	0.03
Trigger: Loneliness	0.117	0.112	29.1	< 0.001	0.34	0.36	0.031	7.38	0.007	−0.18
Trigger: Disclosure	0.061	0.059	14.5	<0.001	0.24	0.049	0.048	11.6	<0.001	−0.22
Trigger: Boredom	0.045	0.040	9.69	0.002	0.20	0.046	0.041	9.86	0.002	−0.21
Trigger: Friendship	0.030	0.021	4.96	0.027	0.15	0.010	0.001	0.27	0.604	−0.04
Trigger: Bad mood	0.084	0.072	19.3	<0.001	0.28	0.025	0.018	4.20	0.042	−0.14

Note. Analyses were conducted with age and gender in model 1, and loneliness/perceived social support in model 2.

Data availability

The data that has been used is confidential.

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